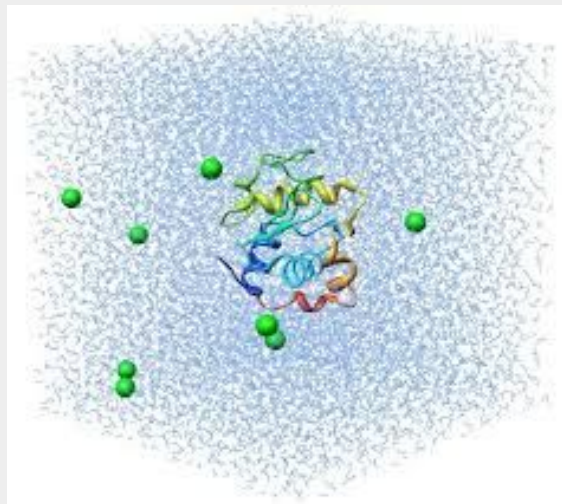


MD Simulation Tutorial using GROMACS



Is there anything static??

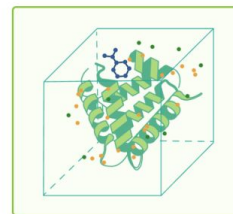
Introduction

- **Molecular Dynamics (MD)** is a computational simulation method that studies the time-dependent behavior of atoms and molecules.
- It calculates how a system evolves by solving **Newton's second law of motion**, $F=ma$, where the force on each particle determines its acceleration.
- By integrating the equations of motion, MD generates a **trajectory** that tracks the positions, velocities, and accelerations of all particles over time.
- From these trajectories, one can determine average **properties** and explore both thermodynamic and kinetic phenomena.
- MD is a **deterministic** method: knowing the initial positions and velocities allows prediction of the system's future state.
- It enables the exploration of macroscopic properties through microscopic simulations.

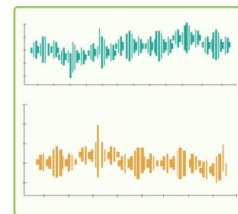
Molecular Dynamics Simulation



Protein-ligand
complex



MD simulation

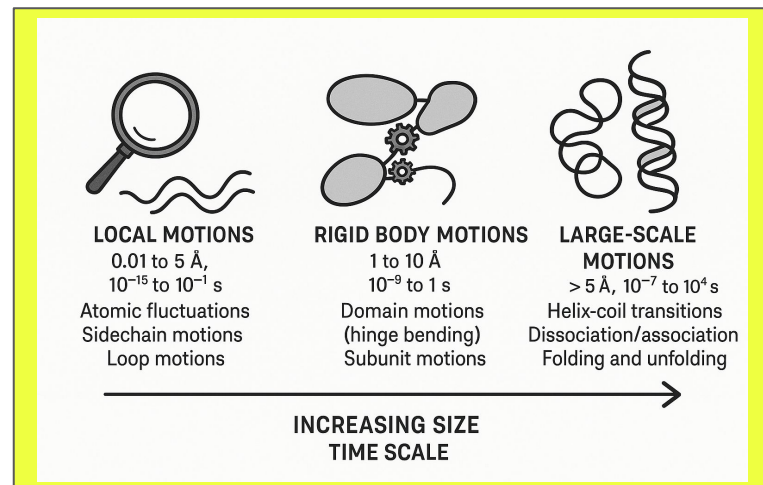


Trajectory analysis



Purpose

- Understanding the physical basis of the structure and function of biological macromolecules.
- Biological molecules exhibit a wide range of time scales over which specific processes occur; for example
 - **Local Motions (0.01 to 5 Å; 10^{-15} to 10^{-1} seconds)** — Atomic fluctuations, side-chain motions, loop motions.
 - **Rigid Body Motions (1 to 10 Å; 10^{-9} to 1 second)** — Domain motions (e.g., hinge bending), subunit motions.
 - **Large-Scale Motions (>5 Å; 10^{-7} to 10^4 seconds)** — Helix-coil transitions, dissociation/association events, folding and unfolding processes.
- **Applications:**
 - Protein folding
 - Protein-ligand interactions
 - Conformational sampling of small molecules
 - Free energy of binding
 - Drug binding and stability
 - Functional mechanisms



The Basic Idea

- Mimic what atoms in do actually in real scenario, assuming a given potential energy function
- Divide time into discrete time steps, no more than a few femtoseconds each ($1\text{fs} = 10^{-15}\text{ Sec}$).
- At each time steps:
Compute the forces acting on each atom, using a mechanical force fields, when atoms moves a bit, record its position and velocity using Newton's laws of motion
- Iterate this process till the end of production run

Verlet algorithm

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i(t), \quad i = 1, 2, 3, \dots, N, \quad \mathbf{r}_i = \text{position vector of the } i\text{th particle and } \mathbf{F}_i \text{ is the vector of forces acting on the } i\text{th particle.}$$

$$\mathbf{r}_i(t + \Delta t) + \mathbf{r}_i(t - \Delta t) = 2\mathbf{r}_i(t) + \ddot{\mathbf{r}}_i(t)\Delta t^2 + O(\Delta t^4).$$

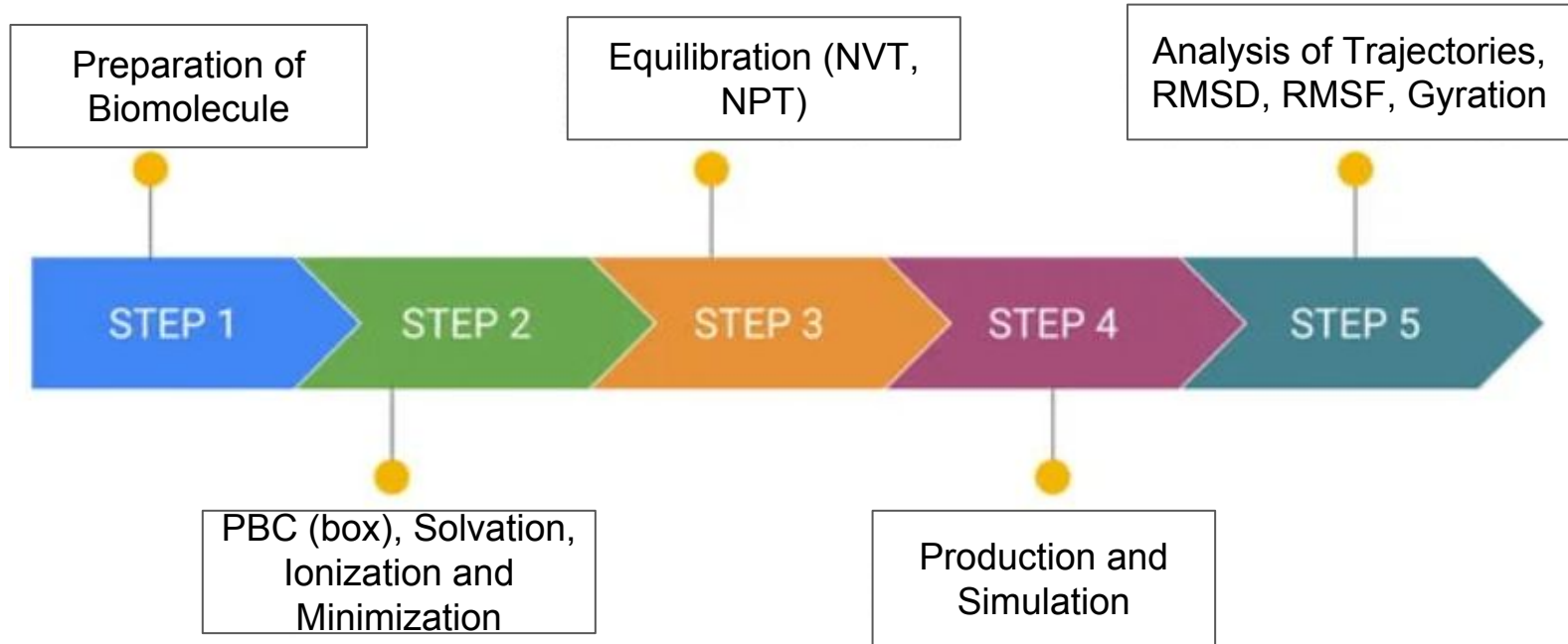
$$\mathbf{r}_i(t + \Delta t) = 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \Delta t) + \frac{1}{m_i} \mathbf{F}_i(t) \Delta t^2 + O(\Delta t^4).$$

Leapfrog verlet Algorithm

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \delta_t \mathbf{v}_{i+1/2}$$

$$\mathbf{v}_{i+1/2} = \mathbf{v}_{i-1/2} + \delta_t F(\mathbf{x}_i) / m$$

Steps



What is a Force Field?

- A force field is a set of equations and parameters to model the potential energy of a molecular system.

Includes terms for:

Bond stretching

Angle bending

Torsional angles (dihedrals)

Non-bonded interactions (van der Waals and electrostatics)

$$\begin{aligned}
 U(R) = & \sum_{bonds} k_r (r - r_{eq})^2 \\
 & + \sum_{angles} k_\theta (\theta - \theta_{eq})^2 \\
 & + \sum_{dihedrals} k_\phi (1 + \cos[n\phi - \gamma]) \\
 & + \sum_{impropers} k_\omega (\omega - \omega_{eq})^2 \\
 & + \sum_{i < j}^{atoms} \epsilon_{ij} \left[\left(\frac{r_m}{r_{ij}} \right)^{12} - 2 \left(\frac{r_m}{r_{ij}} \right)^6 \right] \\
 & + \sum_{i < j}^{atoms} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}
 \end{aligned}$$

bond

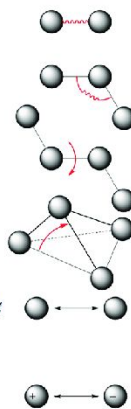
angle

dihedral

improper

van der Waals

electrostatic



Common force fields: AMBER, CHARMM, GROMOS, OPLS.

Step 1: Download & Prepare the Protein Structure

□ **Download** Protein PDB from RCSB PDB

Protein Used: Hen Egg White Lysozyme (PDB ID:
1AKI)

The screenshot displays the RCSB PDB website for the protein structure 1AKI. The page features a 3D ribbon diagram of the protein structure on the left. The main content area on the right provides detailed information about the structure, including its title, DOI, classification, and organism. A dropdown menu is open, showing various download options for the structure. A red arrow points to the 'Ballus gallus' entry in the organism list.

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Versions

1AKI | pdb_00001aki

THE STRUCTURE OF THE ORTHORHOMBIC FORM OF HEN EGG WHITE LYSOZYME AT 1.5 ANGSTROMS RESOLUTION

PDB DOI: <https://doi.org/10.2210/pdb1AKI/pdb>

Classification: **HYDROLASE**

Organism(s): **Ballus gallus**

Mutation(s): No

Deposited: 1997-05-19 Released: 1997-11-19

Deposition Author(s): Carter, D., He, J., Ruble, J.R., Wright, B.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 1.50 Å

Display Files | Download Files | Data API

- FASTA Sequence
- PDBx/mmCIF Format
- PDBx/mmCIF Format (gz)
- BinaryCIF Format (gz)
- Legacy PDB Format
- Legacy PDB Format (gz)
- PDBML/XML Format (gz)
- Structure Factors (CIF)
- Structure Factors (CIF - gz)
- Validation Full PDF

Full Report

Value 6

Preparation

- Visualize
- Check for Missing Atoms
- Remove crystal water molecules

Command

```
grep -v HOH 1aki.pdb >
```

```
HEADER      HYDROLASE                      19-MAY-97   1AKI
TITLE       THE STRUCTURE OF THE ORTHORHOMBIC FORM OF HEN EGG-WHITE LYSOZYME AT
TITLE       2 1.5 ANGSTROMS RESOLUTION
COMPND      MOL_ID: 1;
COMPND      2 MOLECULE: LYSOZYME;
COMPND      3 CHAIN: A;
COMPND      4 EC: 3.2.1.17
SOURCE      MOL_ID: 1;
SOURCE      2 ORGANISM_SCIENTIFIC: GALLUS GALLUS;

ATOM       576  NE  ARG  A  73           18.928  43.771  2.617  1.00 33.74      N
ATOM       577  CZ  ARG  A  73           17.980  44.302  1.823  1.00 33.76      C
ATOM       578  NH1 ARG  A  73           17.841  43.934  0.544  1.00 35.78      N
ATOM       579  NH2 ARG  A  73           17.571  45.545  2.146  1.00 33.83      N
ATOM       580  N   ASN  A  74           24.346  41.143 -0.705  1.00 17.74      N
ATOM       581  CA  ASN  A  74           25.780  40.833 -0.695  1.00 18.87      C

HETATM     1074  O   HOH  A  201         25.892  35.973  11.563  0.46 18.15      O
HETATM     1075  O   HOH  A  202         24.790  25.182  16.063  0.46 17.64      O
HETATM     1076  O   HOH  A  203         12.580  21.214   5.006  0.51 17.97      O
HETATM     1077  O   HOH  A  204         19.687  23.750 -4.851  0.37 18.08      O
HETATM     1078  O   HOH  A  205         27.098  35.956 -12.358  0.39 18.71      O
HETATM     1079  O   HOH  A  206         37.255   9.634  10.002  0.46 18.39      O
HETATM     1080  O   HOH  A  207         43.755  23.843   8.038  0.38 17.96      O

CONNECT     48  981
CONNECT     238 889
CONNECT     513 630
CONNECT     601 724
CONNECT     630 513
CONNECT     724 601
CONNECT     889 238
CONNECT     981 48
MASTER      290   0   0   8   2   0   0   6 1079   1   8  10
END
```

1aki.pdb

b

```
HEADER      HYDROLASE                      19-MAY-97   1AKI
TITLE       THE STRUCTURE OF THE ORTHORHOMBIC FORM OF HEN EGG-WHITE LYSOZYME AT
TITLE       2 1.5 ANGSTROMS RESOLUTION
COMPND      MOL_ID: 1;
COMPND      2 MOLECULE: LYSOZYME;
COMPND      3 CHAIN: A;
COMPND      4 EC: 3.2.1.17
SOURCE      MOL_ID: 1;
SOURCE      2 ORGANISM_SCIENTIFIC: GALLUS GALLUS;

ATOM       576  NE  ARG  A  73           18.928  43.771  2.617  1.00 33.74      N
ATOM       577  CZ  ARG  A  73           17.980  44.302  1.823  1.00 33.76      C
ATOM       578  NH1 ARG  A  73           17.841  43.934  0.544  1.00 35.78      N
ATOM       579  NH2 ARG  A  73           17.571  45.545  2.146  1.00 33.83      N
ATOM       580  N   ASN  A  74           24.346  41.143 -0.705  1.00 17.74      N
ATOM       581  CA  ASN  A  74           25.780  40.833 -0.695  1.00 18.87      C

ATOM       999  CD1  LEU  A  129          39.112  12.657  4.191  1.00 26.43      C
ATOM      1000  CD2  LEU  A  129          37.310  12.325  5.886  1.00 25.15      C
ATOM      1001  OXT  LEU  A  129          43.232  12.675  6.905  1.00 34.20      O
TER        1002  LEU  A  129

CONNECT     48  981
CONNECT     238 889
CONNECT     513 630
CONNECT     601 724
CONNECT     630 513
CONNECT     724 601
CONNECT     889 238
CONNECT     981 48
MASTER      290   0   0   8   2   0   0   6 1079   1   8  10
END
```

1AKI_clean.pdb

b

Step 2: Generate Topology using 'pdb2gmx'

Purpose of pdb2gmx is to generate

Files Generated	File name	Purpose
Topology File	topol.top	System topology
Position Restraints File	posre.itp	Position restraints for heavy atoms
Processed Structure	1AKI_processed.gro	GROMACS-readable structure with hydrogens

Command

```
gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water  
spce
```

Input PDB file

Output coordinate file in .gro format

Specifies the water model (SPC/E) in this case

Other Useful Flags

- “-**ignh**”: Ignore hydrogens in input PDB
- “-**ter**”: Manually assign termini charges

You will be prompted to **select a force field**.

```
Select the Force Field:
```

```
From '/usr/local/gromacs/share/gromacs/top':
```

```
1: AMBER03 protein, nucleic AMBER94 (Duan et al., J. Comp. Chem. 24, 1999-2012, 2003)
2: AMBER94 force field (Cornell et al., JACS 117, 5179-5197, 1995)
3: AMBER96 protein, nucleic AMBER94 (Kollman et al., Acc. Chem. Res. 29, 461-469, 1996)
4: AMBER99 protein, nucleic AMBER94 (Wang et al., J. Comp. Chem. 21, 1049-1074, 2000)
5: AMBER99SB protein, nucleic AMBER94 (Hornak et al., Proteins 65, 712-725, 2006)
6: AMBER99SB-ILDN protein, nucleic AMBER94 (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010)
7: AMBERGS force field (Garcia & Sanbonmatsu, PNAS 99, 2782-2787, 2002)
8: CHARMM27 all-atom force field (CHARM22 plus CMAP for proteins)
9: GROMOS96 43a1 force field
10: GROMOS96 43a2 force field (improved alkane dihedrals)
11: GROMOS96 45a3 force field (Schuler JCC 2001 22 1205)
12: GROMOS96 53a5 force field (JCC 2004 vol 25 pag 1656)
13: GROMOS96 53a6 force field (JCC 2004 vol 25 pag 1656)
14: GROMOS96 54a7 force field (Eur. Biophys. J. (2011), 40,, 843-856, DOI: 10.1007/s00249-011-0700-9)
15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)
```

For this tutorial choose: **15: OPLS-AA/L all-atom force field**

15

Using the Oplsaa force field in directory oplsaaff

going to rename oplsaaff/aminoacids.r2b

Opening force field file /usr/local/gromacs/share/gromacs/top/oplsaa.ff/aminoacids.r2b

Reading 1AKI_clean.pdb...

summary and confirmation of successful topology generation

```
There are 5187 dihedrals, 426 impropers, 3547 angles
          5106 pairs,      1984 bonds and      0 virtual sites
```

```
Total mass 14313.193 a.m.u.
```

```
Total charge 8.000 e
```

```
Writing topology
```

```
Writing coordinate file...
```

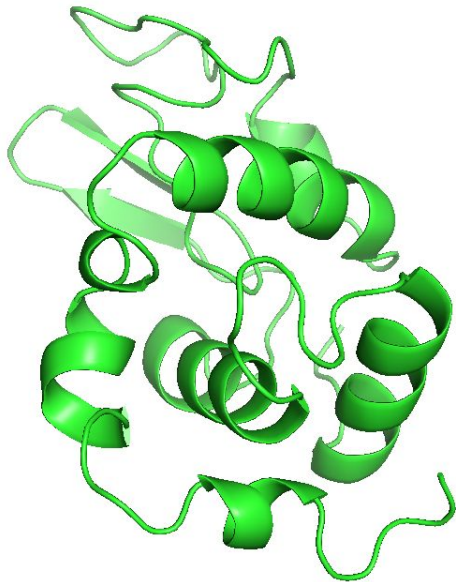
```
----- PLEASE NOTE -----
```

```
You have successfully generated a topology from: 1AKI_clean.pdb.
```

```
The Oplsaa force field and the spce water model are used.
```

```
----- ETON ESAELP -----
```

```
GROMACS reminds you: "Years of calculations and the stress, My science is waiting, nearly complete"
```



1AKI_processed.gro

.gro stands for GROningen molecular dynamics file. It's the default coordinate file format used in GROMACS

```
File "topol.top" was generated
By user: rudrakshi (1004)
On host: iiitd
At date: Thu Apr 10 19:27:08 2025

This is a standalone topology file

Created by:
      :- GROMACS - gmh pdb2gmh, 2023.2 (-)

Executable: /usr/local/gromacs/bin/gmh
Data prefix: /usr/local/gromacs
Working dir: /home/rudrakshi/cadd_md_Tutorial
Command line:
gmh pdb2gmh -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
Force field was read from the standard GROMACS share directory.

; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name          nrexcl
Protein_chain_A  3

[ atoms ]
; nr      type  resnr residue  atom   cgnr   charge   mass  typeB   chargeB   massB
; residue  1 LYS  rtp  LYSH  q +2.0
1  opls_287  1    LYS      N       1      -0.3    14.0027
2  opls_290  1    LYS      H1      1       0.33   1.008
```

```
; Include water topology
#include "oplsaa.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water oxygen
[ position_restraints ]
; i funct  fcx      fcy      fcz
1  1      1000    1000    1000
#endif

; Include topology for ions
#include "oplsaa.ff/ions.itp"

[ system ]
; Name
LYSOZYME

[ molecules ]
; Compound          #mols
Protein_chain_A     1
```

topol.top

The system topology is a file that describes all the components of your system, including atoms, bonds, charges, angles, and interactions, according to a chosen force field. In simple terms it defines how your molecules are built, how they interact, and how they move during the simulation

```
; In this topology include file, you will find position restraint
; entries for all the heavy atoms in your original pdb file.
; This means that all the protons which were added by pdb2gmh are
; not restrained.
```

```
[ position_restraints ]
; atom  type    fx      fy      fz
1       1      1000    1000    1000
5       1      1000    1000    1000
7       1      1000    1000    1000
10      1      1000    1000    1000
13      1      1000    1000    1000
16      1      1000    1000    1000
19      1      1000    1000    1000
```

posre.itp

It contains information used to restrain (keep fixed or limit movement of) certain atoms — usually the heavy atoms (non-hydrogens) of a protein or molecule — during early stages of a molecular dynamics (MD) simulation.

Step 3: Examining the Topology File ('topol.top')

- Calls the OPLS-AA force field parameters/definitions.

```
; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"
```

```
[ moleculetype ]
; Name          nrexcl
Protein_chain_A 3
```

```
[ atoms ]
;  nr      type  resnr residue  atom  cgnr      charge      mass  typeB      chargeB      massB
; residue  1 LYS rtp LYSH q +2.0
   1  opls_287      1   LYS      N      1      -0.3      14.0027
   2  opls_290      1   LYS     H1      1       0.33       1.008
   3  opls_290      1   LYS     H2      1       0.33       1.008
   4  opls_290      1   LYS     H3      1       0.33       1.008
```

- **Molecule Type Definition**

- Protein_chain_A is the molecule **name**
- **nrexcl** = 3: excludes nonbonded interactions within three bonds.

- **Atom-Level Information**, each atom is described by:

- **nr** : Atom number
- **type** : Force field atom type (e.g., `opls\_287`)
- **resnr**, **residue**: Residue number and name
- **atom** : Atom name (from PDB)
- **cgnr** : Charge group number (used for charge grouping)
- **charge**, **mass** : Atom's partial charge and mass
- **typeB**, **chargeB**, **massB** : Used in free energy perturbation (usually ignored)

- **Bonded Interactions,**

Sections include

- [bonds]
- [pairs]
- [angles]
- [dihedrals]

Position Restraints

'posre.itp': Generated during
'pdb2gmx'

Solvent Topology

- Includes parameters for "SPC/E water" (as selected via **-water spce**)
- Other typical choices for water include SPC, TIP3P, and TIP4P.

System-Level Definitions

- system: A label for your system (used in outputs).
- molecules : Lists molecule names and their counts.

Common Errors:

- Mismatched molecule names: Must match coordinate file (.gro) order.
- Wrong order of molecules in [molecules] vs .gro file

```
; Include Position restraint file
#ifdef POSRES
#include "posre.itp"
#endif

; Include water topology
#include "oplsaa.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water oxygen
[ position_restraints ]
; i funct      fcx      fcy      fcz
  1    1      1000      1000      1000
#endif

; Include topology for ions
#include "oplsaa.ff/ions.itp"

[ system ]
; Name
LYSOZYME

[ molecules ]
; Compound      #mols
Protein_chain_A      1
```

Step 4: Define the Simulation Box

Command

```
gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt
```

cubic

Input structure file
(from pdb2gmx).

Centers the protein in the box

Output structure with box dimensions.

Defines a cubic box

Ensures at least 1.0 nm between the protein and the box edge

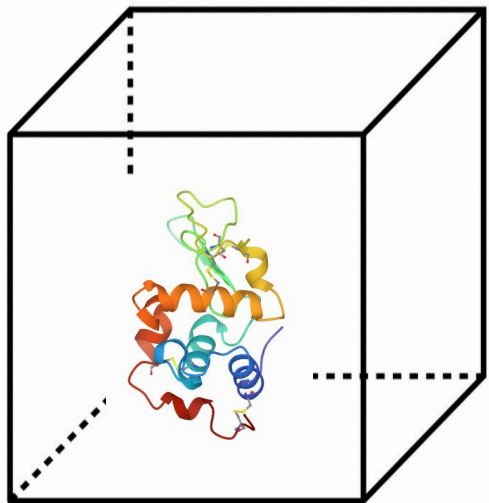
```
Read 1960 atoms
Volume: 123.376 nm^3, corresponds to roughly 55500 electrons
No velocities found
  system size :  3.817  4.234  3.454 (nm)
  diameter    :  5.010                (nm)
  center      :  2.781  2.488  0.017 (nm)
  box vectors :  5.906  6.845  3.052 (nm)
  box angles  :  90.00  90.00  90.00 (degrees)
  box volume  : 123.38                (nm^3)
  shift       :  0.724  1.017  3.488 (nm)
new center    :  3.505  3.505  3.505 (nm)
new box vectors :  7.010  7.010  7.010 (nm)
new box angles :  90.00  90.00  90.00 (degrees)
new box volume : 344.48                (nm^3)

GROMACS reminds you: "Everybody's Good Enough For Some Change" (LIVE)
```

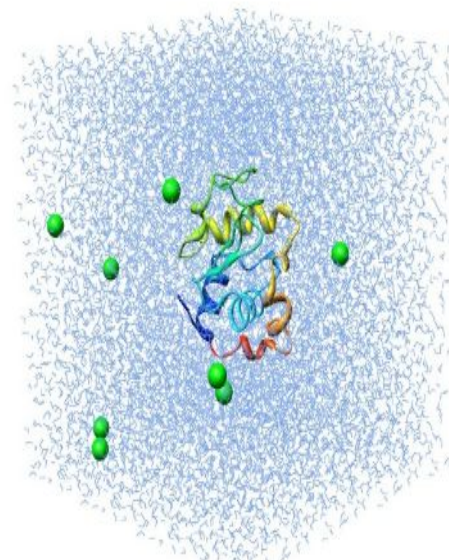

Representation



Force field
applied



Protonated protein in
an aqueous system is
neutralized by the
addition of ions.



Step 5: Add Solvent in box

Command

```
gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p  
topol.top
```

The configuration
from 'editconf'

configuration of the solvent
You can use spc216.gro as the
solvent configuration for SPC,
SPC/E, or TIP3P water, since they
are all three-point water models

Output structure solvated with water

Topology file that gets automatically updated with
the number of added water molecules

```
Output configuration contains 33892 atoms in 10773 residues  
Volume           :    344.484 (nm^3)  
Density           :    997.935 (g/l)  
Number of solvent molecules: 10644
```

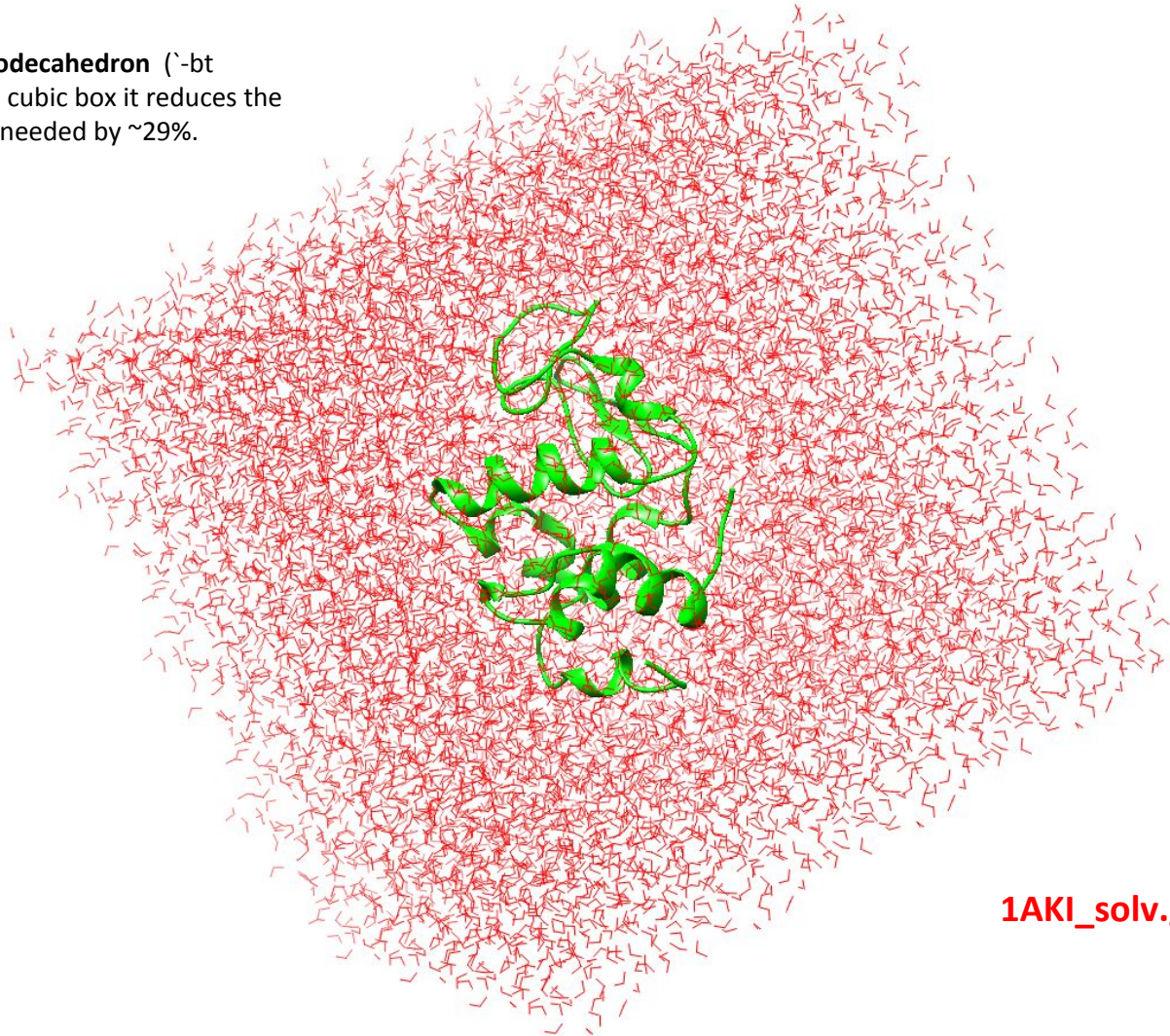
```
Processing topology  
Adding line for 10644 solvent molecules with resname (SOL) to topology file (topol.top)  
Back Off! I just backed up topol.top to ./#topol.top.1#
```

```
; Include topology for ions  
#include "oplsaa.ff/ions.itp"  
  
[ system ]  
; Name  
LYSOZYME in water  
  
[ molecules ]  
; Compound      #mols  
Protein_chain_A    1  
SOL                10644
```

**update in
topol.top**

Pro Tip:

Consider using a **rhombic dodecahedron** (``-bt dodecahedron``) instead of a cubic box it reduces the number of water molecules needed by ~29%.



1AKI_solv.gro

Step 6: Preparing for Ion Addition with grompp

First, you generate the .tpr file, which is a binary input describing your system at the atomic level.

Command

```
gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o
```

ions.tpr

Parameter file

Structure file with
solvent added

Topology file

Output run input file for
use with 'genion

```
; ions.mdp - used as input into grompp to generate ions.tpr
; Parameters describing what to do, when to stop and what to save
integrator      = steep           ; Algorithm (steep = steepest descent minimization)
emtol           = 1000.0          ; Stop minimization when the maximum force < 1000.0 kJ/mol/nm
emstep          = 0.01            ; Minimization step size
nsteps          = 50000           ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate the interactions
nstlist         = 1              ; Frequency to update the neighbor list and long range forces
cutoff-scheme   = Verlet          ; Buffered neighbor searching
ns_type         = grid           ; Method to determine neighbor list (simple, grid)
coulombtype     = cutoff          ; Treatment of long range electrostatic interactions
rcoulomb        = 1.0             ; Short-range electrostatic cut-off
rvdw            = 1.0             ; Short-range Van der Waals cut-off
pbc             = xyz             ; Periodic Boundary Conditions in all 3 dimensions
```

ions.mdp

<http://www.mdtutorials.com/gmx/lysozyme/Files/ions.mdp>

Step 7: Adding Ions with genion

Command

```
gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL
```

-neutral

Input file from
grompp

Output with ions
added

Updated
Topology file

Name of the positive ion

Name of the negative ion

Ensures system neutrality

- During execution, you'll be prompted to choose the group where ions will be placed. Always choose the “SOL” group (usually group 13), so water molecules are replaced—not protein atoms.

```
Will try to add 0 NA ions and 8 CL ions.
Select a continuous group of solvent molecules
Group  0 (      System) has 33892 elements
Group  1 (      Protein) has 1960 elements
Group  2 (    Protein-H) has 1001 elements
Group  3 (      C-alpha) has 129 elements
Group  4 (    Backbone) has 387 elements
Group  5 (    MainChain) has 517 elements
Group  6 ( MainChain+Cb) has 634 elements
Group  7 ( MainChain+H) has 646 elements
Group  8 (    SideChain) has 1314 elements
Group  9 ( SideChain-H) has 484 elements
Group 10 (    Prot-Masses) has 1960 elements
Group 11 (   non-Protein) has 31932 elements
Group 12 (      Water) has 31932 elements
Group 13 (        SOL) has 31932 elements
Group 14 (   non-Water) has 1960 elements
Select a group: 13
Selected 13: 'SOL'
Number of (3-atomic) solvent molecules: 10644

Processing topology
Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.

Back Off! I just backed up topol.top to ./#topol.top.2#
Using random seed 1455283447.
Replacing solvent molecule 460 (atom 3340) with CL
Replacing solvent molecule 10261 (atom 32743) with CL
Replacing solvent molecule 2846 (atom 10498) with CL
Replacing solvent molecule 10125 (atom 32335) with CL
Replacing solvent molecule 6936 (atom 22768) with CL
```

```
; Include topology for ions
#include "oplsaa.ff/ions.itp"

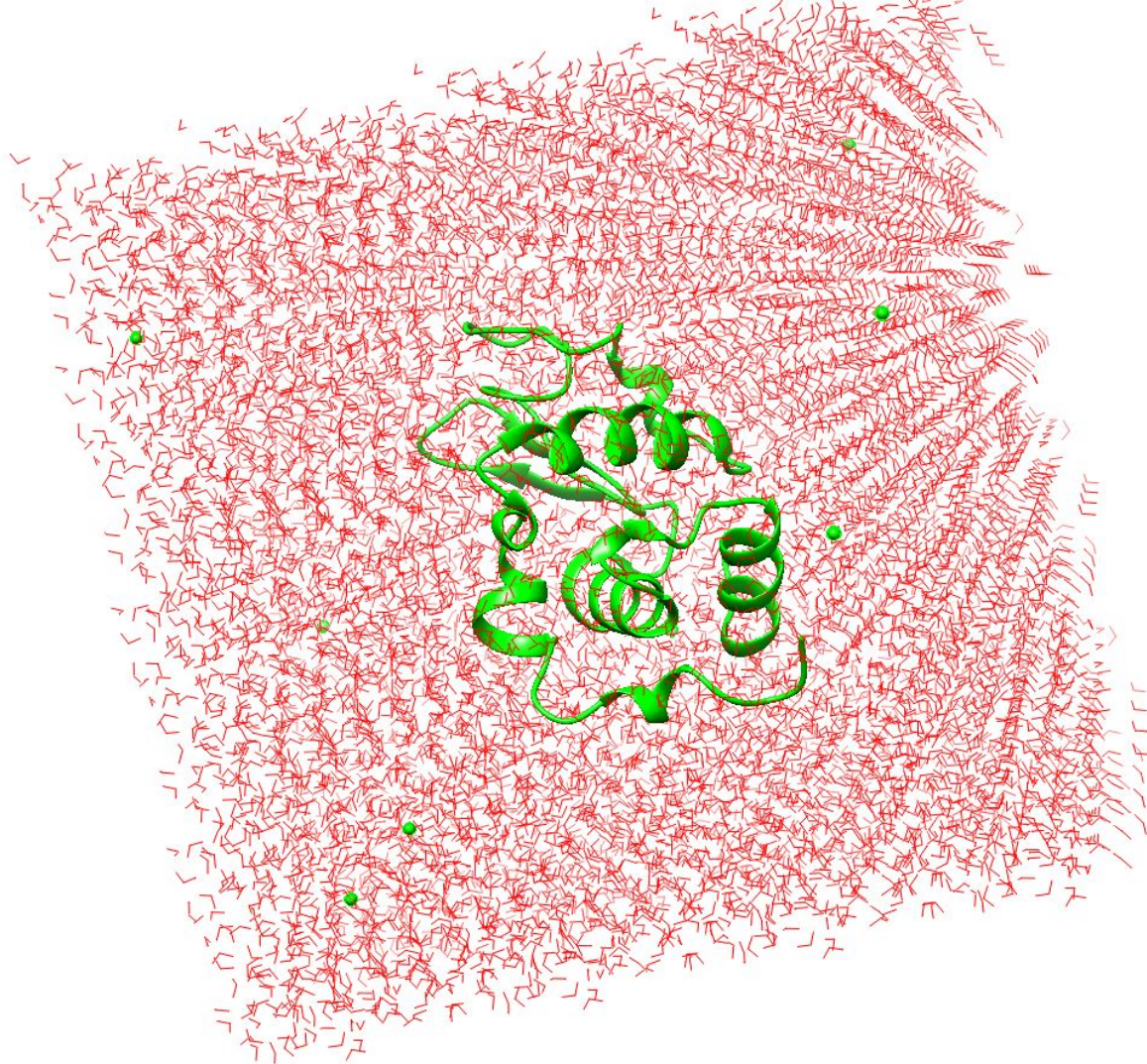
[ system ]
; Name
LYSOZYME in water

[ molecules ]
; Compound      #mols
Protein_chain_A    1
SOL                10636
CL                 8
```

update in
topol.top

Pro Tip:

- You can also use genion to add a specified concentration of ions in addition to simply neutralizing the system by specifying the -neutral and -conc options in conjunction
- You can double-check the final charge of the system with: **gmx check -s ions.tpr**



1AKI_solv_ions.gro

Step 8: Generate .tpr File for Minimization

Command

```
gmxdump -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o
```

em.tpr

Parameter file
Minimization
settings (e.g.,
steepest
descent
integrator,
emtol, etc.)

Structure after
solvation and ion
addition

Topology file

Output binary run file

Troubleshooting Tip : If you get a "number of coordinates does not match topology" error, check if 'topol.top' reflects the correct number of SOL and ions

```
; minim.mdp - used as input into grompp to generate em.tpr
; Parameters describing what to do, when to stop and what to save
integrator = steep           ; Algorithm (steep = steepest descent minimization)
emtol      = 1000.0          ; Stop minimization when the maximum force < 1000.0 kJ/mol/nm
emstep     = 0.01            ; Minimization step size
nsteps     = 50000           ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate the interactions
nstlist    = 1               ; Frequency to update the neighbor list and long range forces
cutoff-scheme = Verlet        ; Buffered neighbor searching
ns_type     = grid           ; Method to determine neighbor list (simple, grid)
coulombtype  = PME           ; Treatment of long range electrostatic interactions
rcoulomb    = 1.0            ; Short-range electrostatic cut-off
rvdw        = 1.0            ; Short-range Van der Waals cut-off
pbc         = xyz            ; Periodic Boundary Conditions in all 3 dimensions
```

minim.mdp

<http://www.md-tutorials.com/gmx/lysozyme/Files/minim.mdp>

Step 9: Run Energy Minimization

Command

```
gmx mdrun -v -deffnm
```

```
em
```

Key Outputs

- **em.gro** : Final energy-minimized structure.
- **em.log** : EM log file.
- **em.edr** : Energy data (used for plotting).
- **em.trr** : Trajectory file (optional but useful for visualizing EM process).

```
Step= 773, Dmax= 8.2e-03 nm, Epot= -5.85964e+05 Fmax= 1.11986e+04, atom= 736
Step= 774, Dmax= 9.8e-03 nm, Epot= -5.86032e+05 Fmax= 3.97253e+03, atom= 736
Step= 776, Dmax= 5.9e-03 nm, Epot= -5.86048e+05 Fmax= 5.10735e+03, atom= 736
Step= 777, Dmax= 7.1e-03 nm, Epot= -5.86065e+05 Fmax= 5.82864e+03, atom= 736
Step= 778, Dmax= 8.5e-03 nm, Epot= -5.86077e+05 Fmax= 7.25194e+03, atom= 736
Step= 779, Dmax= 1.0e-02 nm, Epot= -5.86088e+05 Fmax= 8.48163e+03, atom= 736
Step= 780, Dmax= 1.2e-02 nm, Epot= -5.86091e+05 Fmax= 1.03527e+04, atom= 736
Step= 782, Dmax= 7.3e-03 nm, Epot= -5.86155e+05 Fmax= 9.64136e+02, atom= 736
```

```
writing lowest energy coordinates.
```

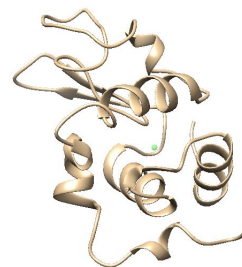
```
Steepest Descents converged to Fmax < 1000 in 783 steps
```

```
Potential Energy = -5.8615500e+05
```

```
Maximum force = 9.6413641e+02 on atom 736
```

```
Norm of force = 2.1687311e+01
```

```
GROMACS reminds you: "I Do It All the Time" (Magnapop)
```



em.gr

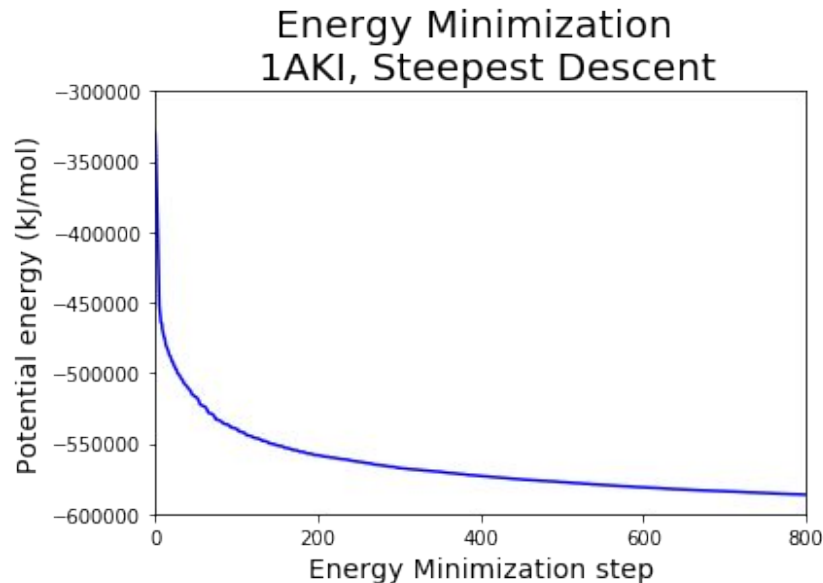
o

Step 10: Optional: Plot Energy Curve

Command

```
gmx energy -f em.edr -o  
potential.xvg
```

```
Select the terms you want from the following list by  
selecting either (part of) the name or the number or a combination.  
End your selection with an empty line or a zero.  
-----  
 1 Bond                2 Angle                3 Proper-Dih.          4 Ryckaert-Bell.  
 5 LJ-14                6 Coulomb-14           7 LJ-(SR)              8 Coulomb-(SR)  
 9 Coul.-recip.         10 Potential           11 Pressure            12 Vir-XX  
13 Vir-XY               14 Vir-XZ              15 Vir-YX              16 Vir-YY  
17 Vir-YZ               18 Vir-ZX              19 Vir-ZY              20 Vir-ZZ  
21 Pres-XX              22 Pres-XY             23 Pres-XZ             24 Pres-YX  
25 Pres-YY              26 Pres-YZ             27 Pres-ZX             28 Pres-ZY  
29 Pres-ZZ              30 #Surf*SurfTen       31 T-rest  
  
10  
0  
Last energy frame read 619 time 782.000  
  
Statistics over 783 steps [ 0.0000 through 782.000 ps ], 1 data sets  
All statistics are over 620 points (frames)  
  
Energy  
-----  
Potential                Average    Err.Est.    RMSD  Tot-Drift  
-----  
Potential                -563429    12000      29448  -77095.6 (kJ/mol)  
  
GROMACS reminds you: "Clickety Click" (System Manager From Hell)
```



This {potential.xvg} plot should show a nice drop in energy as the system stabilizes—something like a downward curve tapering off.

- you'll be prompted to choose a group
Enter: (10 = Potential Energy, 0 to finish
selection)

It will give "potential.xvg" output

Step 11: Run NVT Equilibration (Constant Volume & Temperature)

□ Generate NVT .tpr file

Command

```
gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o  
nvt.tpr
```

```
title           = OPLS Lysozyme NVT equilibration  
define          = -DPOSRES ; position restrain the protein  
; Run parameters  
integrator      = md        ; leap-frog integrator  
nsteps          = 50000    ; 2 * 50000 = 100 ps  
dt              = 0.002    ; 2 fs  
; Output control  
nstxout         = 500      ; save coordinates every 1.0 ps  
nstvout         = 500      ; save velocities every 1.0 ps  
nstenergy       = 500      ; save energies every 1.0 ps  
nstlog          = 500      ; update log file every 1.0 ps  
; Bond parameters  
continuation    = no       ; first dynamics run  
constraint_algorithm = lincs ; holonomic constraints  
constraints      = h-bonds  ; bonds involving H are constrained  
lincs_iter      = 1        ; accuracy of LINCS  
lincs_order     = 4        ; also related to accuracy  
; Nonbonded settings  
cutoff-scheme   = Verlet   ; Buffered neighbor searching  
ns_type         = grid     ; search neighboring grid cells  
nstlist         = 10       ; 20 fs, largely irrelevant with Verlet  
rcoulomb        = 1.0      ; short-range electrostatic cutoff (in nm)  
rvdw            = 1.0      ; short-range van der Waals cutoff (in nm)  
DispCorr        = EnerPres ; account for cut-off vdW scheme  
; Electrostatics  
coulombtype     = PME      ; Particle Mesh Ewald for long-range electrostatics  
pme_order       = 4        ; cubic interpolation  
fourierspacing  = 0.16    ; grid spacing for FFT  
; Temperature coupling is on  
tcoupl          = V-rescale ; modified Berendsen thermostat  
tc-grps         = Protein Non-Protein ; two coupling groups - more accurate  
tau_t           = 0.1      0.1 ; time constant, in ps  
ref_t           = 300      300 ; reference temperature, one for each group, in K  
; Pressure coupling is off  
pcoupl          = no       ; no pressure coupling in NVT  
; Periodic boundary conditions  
pbc             = xyz      ; 3-D PBC  
; Velocity generation  
gen_vel         = yes      ; assign velocities from Maxwell distribution  
gen_temp        = 300      ; temperature for Maxwell distribution  
gen_seed        = -1       ; generate a random seed
```

nvt.mdp

<http://www.mdtutorials.com/gmx/lysozyme/Files/nvt.mdp>

□ Run the NVT simulation

Command

```
gmx mdrun -deffnm
```

```
nvt
```

```
starting mdrun 'LYSOZYME in water'  
50000 steps, 100.0 ps.
```

```
Writing final coordinates.
```

	Core t (s)	Wall t (s)	(%)
Time:	805.441	40.282	1999.5
	(ns/day)	(hour/ns)	
Performance:	214.490	0.112	

This will generate:

- **nvt.trr** : Trajectory
- **nvt.edr** : Energy file
- **nvt.gro** : Structure after NVT
- **nvt.log** : Log file
- **nvt.cpt** : Checkpoint (very useful for continuing runs)

Step 12: Check Temperature Stability

Command

```
gmx energy -f nvt.edr -o
```

```
temperature.xvg
```

```
Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
```

```
-----
 1 Bond                2 Angle                3 Proper-Dih.          4 Ryckaert-Bell.
 5 LJ-14               6 Coulomb-14           7 LJ-(SR)              8 Disper.-corr.
 9 Coulomb-(SR)        10 Coul.-recip.        11 Position-Rest.     12 Potential
13 Kinetic-En.         14 Total-Energy        15 Conserved-En.     16 Temperature
17 Pres.-DC            18 Pressure            19 Constr.-rmsd       20 Vir-XX
21 Vir-XY              22 Vir-XZ              23 Vir-YX              24 Vir-YY
25 Vir-YZ              26 Vir-ZX              27 Vir-ZY              28 Vir-ZZ
29 Pres-XX             30 Pres-XY             31 Pres-XZ             32 Pres-YX
33 Pres-YY             34 Pres-YZ             35 Pres-ZX             36 Pres-ZY
37 Pres-ZZ             38 #Surf*SurfTen       39 T-Protein           40 T-non-Protein
41 Lamb-Protein        42 Lamb-non-Protein
```

```
16 0
```

```
Last energy frame read 100 time 100.000
```

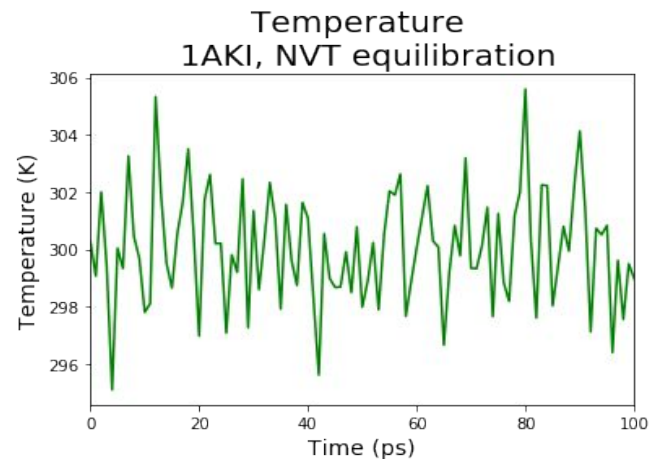
```
Statistics over 50001 steps [ 0.0000 through 100.0000 ps ], 1 data sets
All statistics are over 501 points
```

```
-----
Energy                Average    Err.Est.      RMSD  Tot-Drift
-----
Temperature            299.767      0.21      2.98011  0.627528 (K)
```

- you'll be prompted to choose a group

Enter: (16 = usually corresponds to Temperature
, 0 to finish selection)

It will give "temperature.xvg" output



Your plot should show:

- A quick rise to ~300 K. Then, a stable plateau

If your temperature fluctuates too wildly or doesn't stabilize, you may need to:

- Increase the equilibration time in nvt.mdp
- Check for bad contacts or unstable ions.

Step 13: Run NPT Equilibration

□ Generate NVT .tpr file

Command

```
gmxd grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o  
npt.tpr
```

The -t nvt.cpt flag is crucial: this checkpoint file contains all velocities and simulation state info from the NVT phase.

```
title           = OPLS Lysozyme NPT equilibration  
define         = -DPOSRES ; position restrain the protein  
; Run parameters  
integrator      = md       ; leap-frog integrator  
nstxout        = 50000     ; 2 * 50000 = 100 ps  
nstvout        = 500       ; save coordinates every 1.0 ps  
nstenergy      = 500       ; save velocities every 1.0 ps  
nstlog         = 500       ; save energies every 1.0 ps  
; Output control  
nstxout        = 500       ; update log file every 1.0 ps  
nstvout        = 500       ; save coordinates every 1.0 ps  
nstenergy      = 500       ; save velocities every 1.0 ps  
nstlog         = 500       ; save energies every 1.0 ps  
; Bond parameters  
continuation   = yes       ; Restarting after NVT  
constraint_algorithm = lincs ; holonomic constraints  
constraints    = h-bonds   ; bonds involving H are constrained  
lincs_iter     = 1         ; accuracy of LINCS  
lincs_order    = 4         ; also related to accuracy  
; Nonbonded settings  
cutoff-scheme  = Verlet    ; Buffered neighbor searching  
ns_type        = grid      ; search neighboring grid cells  
nstlist        = 10       ; 20 fs, largely irrelevant with Verlet scheme  
rcoulomb        = 1.0      ; short-range electrostatic cutoff (in nm)  
rvdw            = 1.0      ; short-range van der Waals cutoff (in nm)  
DispCorr       = EnerPres  ; account for cut-off vdW scheme  
; Electrostatics  
coulombtype     = PME       ; Particle Mesh Ewald for long-range electrostatics  
pme_order       = 4         ; cubic interpolation  
fourierspacing = 0.16      ; grid spacing for FFT  
; Temperature coupling is on  
tcoupl          = V-rescale ; modified Berendsen thermostat  
tc-grps         = Protein Non-Protein ; two coupling groups - more accurate  
tau_t           = 0.1      0.1 ; time constant, in ps  
ref_t           = 300      300 ; reference temperature, one for each group, in K  
; Pressure coupling is on  
pcoupl          = Parrinello-Rahman ; Pressure coupling on in NPT  
pcoupltype      = isotropic  ; uniform scaling of box vectors  
tau_p           = 2.0       ; time constant, in ps  
ref_p           = 1.0       ; reference pressure, in bar  
compressibility = 4.5e-5     ; isothermal compressibility of water, bar^-1  
refcoord_scaling = com       ;  
; Periodic boundary conditions  
pbc             = xyz       ; 3-D PBC  
; Velocity generation  
gen_vel         = no        ; Velocity generation is off
```

npt.mdp

<http://www.mdtutorials.com/gmx/lysozyme/Files/npt.mdp>

□ Run the NPT simulation

Command

```
gmx mdrun -deffnm  
npt
```

volume change so pressure and system density can equilibrate to reflect realistic, experimental conditions.

Key Differences in the .mdp for NPT:

- continuation = yes: Continues from NVT. You keep the atomic velocities instead of randomizing them again.
- gen_vel = no : Reuses velocities from NVT instead of generating new ones.
- pcoupl = Parrinello-Rahman: Activates pressure coupling via a barostat — allows the box to change size.

```
starting mdrun 'LYSOZYME in water'  
50000 steps,    100.0 ps.  
  
Writing final coordinates.  
  
                Core t (s)   Wall t (s)         (%)  
      Time:      2463.784     123.199     1999.8  
                (ns/day)    (hour/ns)  
Performance:      70.132       0.342  
  
GROMACS reminds you: "I Need a Little Poison" (Throwing Muses)
```


Step 14: Check Pressure and Density Stability

□ Run the NPT simulation

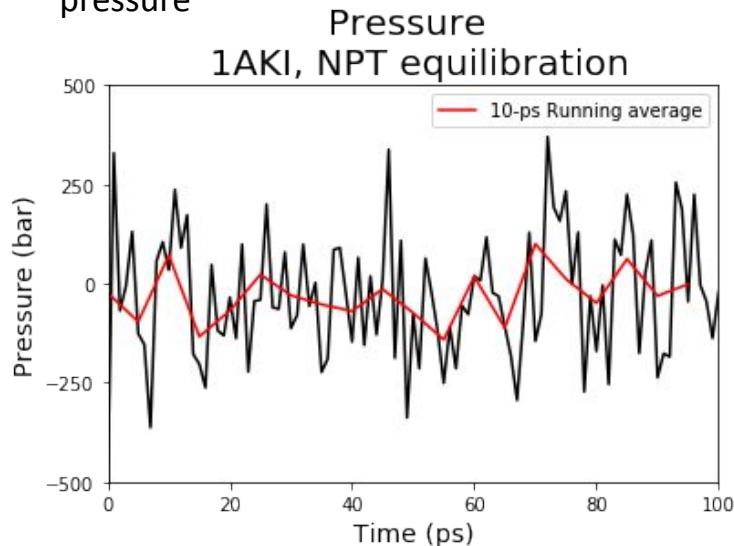
Command

```
gmx energy -f npt.edr -o
```

pressure.xvg

Select option **18** for

pressure



□ Run the NPT simulation

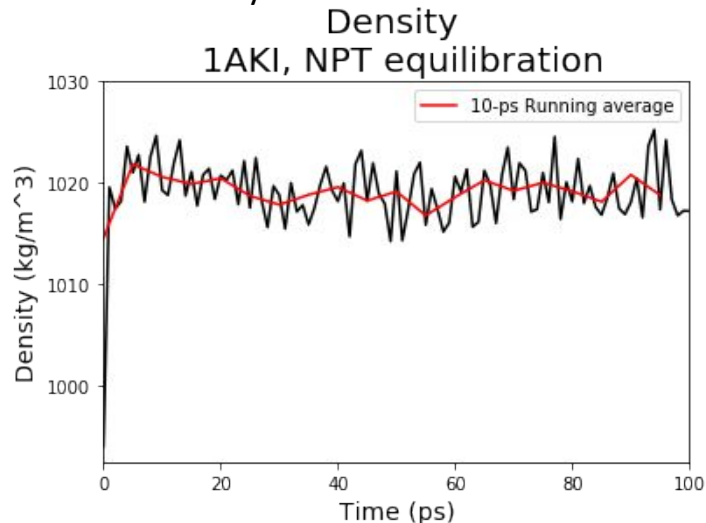
Command

```
gmx energy -f npt.edr -o
```

density.xvg

Select option **24** for

density.



Step 15: production MD

□ Preparation Command (grompp)

Command

```
gmX grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o
```

```
md_0_1.tpr
```

□ Running the Simulation

Command

```
gmX mdrun -deffnm md_0_1 -nb
```

gpu

What's Happening in Production MD?

- No restraints: At this point, atoms are free to move naturally, guided by physical forces.
- Duration: Typically starts with **1 ns** (as in your case), but for biologically relevant insights, simulations often extend to tens or hundreds of ns.
- Ensemble: You're still under NPT conditions (constant Pressure and Temperature).

```
title = OPLS Lysozyme NPT equilibration
; Run parameters
integrator = md ; leap-frog integrator
nstps = 50000 ; 2 * 50000 = 1000 ps (1 ns)
dt = 0.002 ; 2 fs
; Output control
nstxout = 0 ; suppress bulky .trr file by specifying
nstvout = 0 ; 0 for output frequency of nstxout,
nstfout = 0 ; nstfout, and nstfout
nstenergy = 5000 ; save energies every 10.0 ps
nstlog = 5000 ; update log file every 10.0 ps
nstxout-compressed = 5000 ; save compressed coordinates every 10.0 ps
compressed-x-grps = System ; save the whole system
; Bond parameters
continuation = yes ; Restarting after NPT
constraint_algorithm = lincs ; holonomic constraints
constraints = h-bonds ; bonds involving H are constrained
lincs_iter = 1 ; accuracy of LINCS
lincs_order = 4 ; also related to accuracy
; Neighborsearching
cutoff-scheme = Verlet ; Buffered neighbor searching
ns_type = grid ; search neighboring grid cells
nstlist = 10 ; 20 fs, largely irrelevant with Verlet scheme
rcoulomb = 1.0 ; short-range electrostatic cutoff (in nm)
rvdw = 1.0 ; short-range van der Waals cutoff (in nm)
; Electrostatics
coulombtype = PME ; Particle Mesh Ewald for long-range electrostatics
pme_order = 4 ; cubic interpolation
fourierspacing = 0.16 ; grid spacing for FFT
; Temperature coupling is on
tcoupl = V-rescale ; modified Berendsen thermostat
tc-grps = Protein Non-Protein ; two coupling groups - more accurate
tau_t = 0.1 0.1 ; time constant, in ps
ref_t = 300 300 ; reference temperature, one for each group, in K
; Pressure coupling is on
pcoupl = Parrinello-Rahman ; Pressure coupling on in NPT
pcoupltype = isotropic ; uniform scaling of box vectors
tau_p = 2.0 ; time constant, in ps
ref_p = 1.0 ; reference pressure, in bar
compressibility = 4.5e-5 ; isothermal compressibility of water, bar^-1
; Periodic boundary conditions
pbc = xyz ; 3-D PBC
; Dispersion correction
DispCorr = EnerPres ; account for cut-off vdW scheme
; Velocity generation
gen_vel = no ; Velocity generation is off
```

md.mdp

<http://www.md-tutorials.com/gmx/lysozyme/Files/md.mdp>

Analysis

□ Fixing Periodicity with trjconv

Proteins may "jump" across periodic boundaries during simulation, making them look broken or disconnected. You don't want that in your analysis visuals or calculations. To fix that

```
gmx trjconv -s md_0_1.tpr -f md_0_1.xtc -o md_0_1_noPBC.xtc -pbc mol -center
```

Select 1 ("Protein") as the group to be centered and 0 ("System") for output. We will conduct all our analyses on this "corrected" trajectory.

Analysis Type	Command Example	
RMSD	gmx rms -s md_0_1.tpr -f md_0_1_noPBC.xtc -o rmsd.svg -tu ns	Choose 4 ("Backbone") for both the least-squares fit and the group for RMSD calculation
RMSD relative to the crystal structure	gmx rms -s em.tpr -f md_0_1_noPBC.xtc -o rmsd_xtal.svg -tu ns	Choose 4 ("Backbone") for both the least-squares fit and the group for RMSD calculation
RMSF	gmx rmsf -s md_0_1.tpr -f md_0_1_noPBC.xtc -o rmsf.svg -res -oq bfac.pdb (Open bfac.pdb in chimera/pymol - color by bfactor)	Select group 4 ("Backbone") for root mean square calculation
Radius of Gyration	gmx gyrate -s md_0_1.tpr -f md_0_1_noPBC.xtc -o gyrate.svg	Choose group 1 (Protein) for analysis.
H-bond analysis	gmx hbond -f md_0_1_noPBC.xtc -s md_0_1.tpr -num h2bond.svg -g h2bond.log -contact	Specify 2 groups to analyze: Select a group: 1(Protein) Select a group: 13(UNL)

Interested to work on?

De novo protein design

Protein folding mechanism in eukaryotes

Protein Function Prediction (June 1, 2026)

(You will build a model that predicts what a protein does based on its amino acid sequence)

Biomolecular structure Prediction

References

1. Theory of Molecular Dynamics : https://embnet.vitalit.ch/MD_tutorial/pages/MD.Part1.html
2. Introduction to Molecular Dynamics : <https://udel.edu/~arthij/MD.pdf>
3. A very good explanation of workflow : <https://charmm-gui.org/?doc=lecture&module=pdb&lesson=1>
4. Tutorial to be followed : Lysozyme in water
<http://www.mdtutorials.com/gmx/lysozyme/index.html>
A detailed explanation of the same
https://events.prace-ri.eu/event/455/attachments/369/553/MD_Lysozyme.pdf

A 3D molecular structure is visible in the background, featuring various colored ribbons (green, yellow, blue, red) representing different parts of a protein or complex. The structure is partially obscured by a central blue box.

Thank
You
for
your
Patience!